$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{30-24-4}{\sqrt{50}\sqrt{101}} = \frac{\sqrt{2}}{5\sqrt{101}} \Rightarrow \theta = \cos^{-1}\left[\frac{\sqrt{2}}{5\sqrt{101}}\right].$$

Example 8.17

Find the projection of $\overrightarrow{AB}$ on $\overrightarrow{CD}$ where *A, B, C, D* are the points (4, – 3, 0), (7, – 5, – 1), (– 2, 1, 3), (0, 2, 5).

Solution

Let *O* be the origin.

Therefore, $\overrightarrow{OA} = 4\hat{i} - 3\hat{j}$; $\overrightarrow{OB} = 7\hat{i} - 5\hat{j} - \hat{k}$; $\overrightarrow{OC} = -2\hat{i} + \hat{j} + 3\hat{k}$; $\overrightarrow{OD} = 2\hat{j} + 5\hat{k}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 3\hat{i} - 2\hat{j} - \hat{k}$; $\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = 2\hat{i} + \hat{j} + 2\hat{k}$

Projection of $\overrightarrow{AB}$ on $\overrightarrow{CD} = \frac{\overrightarrow{AB}\cdot\overrightarrow{CD}}{|\overrightarrow{CD}|} = \frac{6-2-2}{3} = \frac{2}{3}$.

Example 8.18

If $\vec{a}, \vec{b},$ and $\vec{c}$ are three unit vectors satisfying $\vec{a} - \sqrt{3}\,\vec{b} + \vec{c} = \vec{0}$ then find the angle between $\vec{a}$ and $\vec{c}$.

Solution

Let θ be the angle between $\vec{a}$ and $\vec{c}$.

$$\vec{a} - \sqrt{3}\,\vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \quad |(\vec{a}+\vec{c})| = |\sqrt{3}\,\vec{b}|$$

$$\Rightarrow \quad |\vec{a}|^2 + |\vec{c}|^2 + 2|\vec{a}||\vec{c}|\cos\theta = 3|\vec{b}|^2$$

$$\Rightarrow \quad 1 + 1 + (2)(1)(1)\cos\theta = 3(1)$$

$$\Rightarrow \quad \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}.$$

Example 8.19

Show that the points (4, – 3, 1), (2, – 4, 5) and (1, – 1, 0) form a right angled triangle.

Solution

Trivially they form a triangle. It is enough to prove one angle is $\frac{\pi}{2}$. So find the sides of the triangle.

Let *O* be the point of reference and *A,B,C* be (4, – 3, 1), (2, – 4, 5) and (1, – 1, 0) respectively.

$\overrightarrow{OA} = 4\hat{i} - 3\hat{j} + \hat{k}$, $\overrightarrow{OB} = 2\hat{i} - 4\hat{j} + 5\hat{k}$, $\overrightarrow{OC} = \hat{i} - \hat{j}$

Now, $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = -2\hat{i} - \hat{j} + 4\hat{k}$

Similarly, $\overrightarrow{BC} = -\hat{i} + 3\hat{j} - 5\hat{k}$; $\overrightarrow{CA} = 3\hat{i} - 2\hat{j} + \hat{k}$

Clearly, $\overrightarrow{AB}\cdot\overrightarrow{CA} = 0$

Thus one angle is $\frac{\pi}{2}$. Hence they form a right angled triangle.

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